

SkillForge – Milestone 2 Status Report

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Submitted By: Shanmukh Sahukari

Overview

The Second milestone of the SkillForge full-stack learning platform, focusing on the User Authentication Module, has been successfully completed. This foundational phase established secure user registration and login functionalities, integrated front-end and backend systems, and set up reliable JWT-based authorization.

Progress

- **Frontend:**

Designed and implemented user-friendly Login and Registration pages using HTML, CSS, and JavaScript, fully integrated with backend APIs.

- **Backend:**

Configured a Node.js and Express server, connected to a MySQL database, and developed secure APIs for registration and login with password hashing and JWT token generation.

- **Database:**

Created the necessary database and users table with data insertion verified through Postman.

- **Authentication:**

Implemented JWT middleware for token validation and a protected route to retrieve loggedin user details.

- **Testing:**

Thorough API testing was conducted with Postman to verify registration, login, token authorization, and protected route functionality.

- **Documentation and Repository:**

Completed product and sprint backlogs, requirement analysis, architecture and use case diagrams, and maintained a GitHub repository with all code and documentation.

Challenges

- **CORS issues:**

Resolved by enabling CORS middleware on the server.

- **JSON parsing errors:**

Fixed by adding `express.json()` middleware.

- **JWT token validation failures:**

Improved middleware logic to ensure robust authentication.

- **MySQL connection problems:**

Addressed by verifying credentials and reconnecting the database.

- **UI redirect problems:**

Corrected by updating `window.location.href` logic for seamless navigation.

Next Steps

- Begin development of the Dashboard module to empower user interaction postlogin.
- Implement the Courses module for course browsing and enrollment.
- Develop the Instructor module to support course creation and management.
- Integrate AI-based recommendation features to personalize learning paths.

Conclusion

Milestone 2 has laid a solid foundation by delivering a fully functional and secure authentication system. The successful integration, testing, and documentation of components ensure readiness for upcoming phases of SkillForge development. The team is well-positioned to proceed with the next modules, driving the platform towards its full.